

Name and Surname: _____

Student ITS No: _____

Qualification: _____ Year of Study: _____ Semester: _____

Assignment due date: _____ Date submitted: _____

QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion. (10 Marks)

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document.
Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents

Introduction

Main Body (with relevant subheadings)

Conclusion

References

Submit the assignment in PDF format on Moodle.

Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: _____

Date: _____

QUESTION 1**(30 MARKS)**

1.1 Richfield Consulting recently hired you as a graduate database developer. They're upgrading their data storage from a manual paper-based system to a digital DBMS. As a graduate developer, critically discuss the difference between a paper-based system and a digital DBMS, furthermore, highlight the components, functions as well as the pros and cons of a DBMS.

QUESTION 2**(30 MARKS)**

2.1 Provide the necessary SQL code to create a table named Student, with columns as shown in

Student_No	Student_Name	Student_Surname	Student_Specialisation
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Figure 1. Ensure that the columns have appropriate data types.

(5)

Figure 1: SQL Table

2.2 Provide the necessary SQL code to insert the data as indicated in Figure 2

(5)

	Student_No	Student_Name	Student_Surname	Student_Specialisation
1	1224	Peter	Josheps	DMV
2	1225	Nick	Saint	PRO
3	1226	Mpho	Mchunu	NET
4	1227	Thabisa	van Niekerk	BAS
5	1228	Deon	Holmes	BMG

Figure 2: SQL Table with data

2.3 Provide the necessary SQL code to insert a record with your name, surname, 9999 as the student

	Student_No	Student_Name	Student_Surname	Student_Specialisation
1	1224	Peter	Josheps	DMV
2	1225	Nick	Saint	PRO
3	1226	Mpho	Mchunu	NET
4	1227	Thabisa	van Niekerk	BAS
5	1228	Deon	Holmes	BMG
6	9999	Your Name	Your Surname	DBS

number, and DBS as the specialisation.

(5)

Figure 3: SQL Table with custom data

2.4 Provide the necessary SQL code to change the Student_Surname for Thabisa from van Niekerk to

	Student_No	Student_Name	Student_Surname	Student_Specialisation
1	1224	Peter	Josheps	DMV
2	1225	Nick	Saint	PRO
3	1226	Mpho	Mchunu	NET
4	1227	Thabisa	Ndlovu	BAS
5	1228	Deon	Holmes	BMG
6	9999	Your Name	Your Surname	DBS

Ndlovu.

(5)

Figure 4: SQL Table with updated values

2.5 Provide the necessary SQL Code to change the Column name 'Student_Specialisation' to Student Module (5)

2.6 Provide the necessary SQL code to delete the Student table as well as the database the table is stored in. For this question, assume the Student table is stored in a database named School. (5)

QUESTION 3

(30 MARKS)

3.1 TechGen is upgrading their database from an excel spreadsheet to an RDMBS. They have given you the following business rules. A store can have many employees but only one manager. An employee must be a Technical Assistant or Desktop Technician. A manager can be a store manager

or a departmental manager. There can only be one store in one location, and each store can have up to three (3) departments only.

Draw the EERD for TechGen and indicate multiplicity / cardinality (where applicable) as well as the common attributes in each table. (15)

3.2 Consider the following report below. Convert the report into a set of BCNF tables (if possible). Fully describe each step and clearly show relation schema.

STUD_NO	STUD_NAME	STUD_SNAME	QUAL_ID	QUAL_DESC	FINAL_PROJ	PROJ_NAME
1125	Marco	Polo	1	Degree, Undergrad	Required	3 rd Year Project
5541	Jason	Statham	2	Higher Certificate	Not Required	
6687	Ben	Grimm	1	Degree, Undergrad	Required	3 rd Year Project
2551	Peter	Parker	3	Diploma	Required	3 rd Year Project
6669	Thando	Wolff	5	Honours	Required	Research project
8879	Theodore	Jacobs	8	MBA	Optional	

(15)

*****END*****